

Plugin B: "Vintage Tape Delay"

An analog-modeled echo plugin.

GUI Features

- **Controls:** Knobs for Delay Time, Feedback, and Mix (Dry/Wet).

Architecture & Technical Highlights

AudioProcessorValueTreeState

Management of all parameters via JUCE's professional APVTS system. This guarantees absolute thread safety during data exchange between the audio thread and the operating system's GUI thread.

Ring Buffer (Circular Buffer)

Implementation of a dynamically allocated memory array in RAM via `juce::AudioBuffer`, where the write and read heads run continuously in a circle to cache audio data with zero latency.

Fractional Delay / Linear Interpolation

To prevent "zipper noise" when moving the time control, the read head allows fractional positions. The plugin reads the two actual adjacent samples from RAM and crossfades them seamlessly.

Parameter Smoothing

The pointer does not jump abruptly but glides towards the target value at a rate of 0.1% per sample. This perfectly simulates the physical inertia of an analog tape machine motor, creating the popular, organic pitch-shifting effect during time changes.